..... Massive Crowd Management in Urban Pedestrian Zones

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***Abstract*—This paper presents a bi-level optimization framework for crowd management in large pedestrian zones, integrating anisotropic geometric guidance with discrete directional constraints. The lower level employs a Hughes-type macroscopic crowd model enhanced with metric tensor fields for channel guidance and constrained Hamilton-Jacobi-Bellman (HJB) equations for one-way channel control. The upper level optimizes channel direction configurations and guidance intensities using a Structure-Aware Hybrid Block Optimization (SA-HBO) algorithm. Three raw physical metrics are introduced for efficiency, safety, and load balancing, and are further converted into dimensionless indicators for cross-scheme comparison and scalarization. The proposed method exploits the local response characteristics of Bellman potential fields and employs surrogate pre-screening with Bellman warm-start strategies to reduce computational cost. Numerical experiments demonstrate the effectiveness of the proposed approach.**

***Index Terms*—Crowd dynamics, macroscopic modeling, channel optimization, Hamilton-Jacobi-Bellman equation, bi-level optimization.**

I. Introduction

LARGE-SCALE pedestrian management in urban areas such as waterfront promenades, stadiums, and transportation hubs presents significant challenges for public safety and operational efficiency. The dynamics of dense crowds involve complex interactions between individual behaviors, geometric constraints, and collective motion patterns that are difficult to predict and control.

Macroscopic crowd models, pioneered by Hughes [1], treat pedestrian flows as continuous media described by density and velocity fields. These models offer computational efficiency and analytical tractability compared to microscopic approaches, making them suitable for real-time management applications. However, classical Hughes models assume isotropic movement and cannot adequately represent anisotropic channel geometries or directional control measures such as one-way corridors.

To address these limitations, this paper develops an enhanced modeling framework that integrates:

- Anisotropic geometric guidance via spatially varying metric tensors to represent channel preferences;
- Discrete directional constraints via constrained HJB equations to enforce one-way rules;
- A bi-level optimization structure that treats the simulation as an evaluation oracle for control decisions.

The optimization variables include discrete channel direction configurations and continuous guidance intensity parameters. To handle the mixed discrete-continuous structure and expensive simulation evaluations, we propose a Structure-Aware Hybrid Block Optimization (SA-HBO) framework that exploits problem-specific structures including local Bellman response characteristics and surrogate pre-screening.

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II. Related Work

A. Massive Crowd Management and Control

Crowd management in urban centers has been extensively studied from control-theoretic and optimization perspectives, aiming to enhance safety and efficiency through boundary regulation, facility layout, and intelligent decision-making [5], [6].

In control-theoretic approaches, Zhong et al. [7] designed boundary control strategies for urban traffic flow networks to ensure convergence to uncongested states under disturbances, with stability analysis based on Lyapunov methods [8]. For one-dimensional crowd dynamics, Wadoo and Kachroo [9] employed advection and diffusion control to prevent blocking and shock waves during evacuation. Addressing multi-directional disturbance propagation, Qin [10] proposed a unit sliding mode controller based on integral barrier Lyapunov functions to ensure boundedness of internal state variables, demonstrating strong disturbance rejection capabilities. For heterogeneous corridor regulation, Zhu et al. [11] combined policy learning with neural networks to learn optimal feedback controllers that adjust inflow rates and free-flow speeds, achieving balanced density distribution and reduced congestion.

Physical infrastructure regulation has also proven effective. Studies have shown that strategically placed obstacles can significantly improve outflow, velocity, and local density in merging areas [12], [13]. Yang et al. [14], [15] employed Model Predictive Control (MPC) to optimize barrier lengths at subway bottlenecks, later extending control variables to entrance limiters, gates, and escalator directions for dynamic, real-time crowd regulation. For evacuation guidance, Carmona and Paricio-Garcia [16] proposed dynamic exit-choice recommendations using multinomial Logit models. However, their approach relies on discrete speed instructions (e.g., "stop," "walk," "run") transmitted via electronic wristbands [17], which poses implementation challenges in open urban scenarios.

Recently, intelligent optimization and deep reinforcement learning have emerged as promising alternatives. Liao et al. [18] formulated fence layout as a subset selection problem, while subsequent work [19] proposed a dynamic crowd inflow control system using LSTM networks to capture temporal features and Proximal Policy Optimization (PPO) [20] for real-time entrance flow regulation. Differential Evolution (DE) [21] has also been applied to optimize guardrail layouts and flow guidance in real-world scenarios such as Chengdu East Railway Station, using average travel time and crowd pressure as multi-objective criteria [22].

B. Macroscopic Crowd Modeling

Macroscopic crowd models treat pedestrian flows as continuous media, offering computational efficiency compared to microscopic approaches. Hughes [2] pioneered the continuum theory for pedestrian flow, introducing the classical model where pedestrians move in the direction of steepest descent of a potential function representing remaining travel time to exits.

Extensions of the Hughes framework have incorporated various control and guidance mechanisms. Anisotropic extensions using metric tensors have been developed to represent channel geometries and directional preferences [3], [4]. These models allow for continuous geometric guidance without explicit prohibition, enabling pedestrians to deviate from preferred directions when necessary.

Constrained Hamilton-Jacobi-Bellman (HJB) formulations have been employed to model strict directional constraints such as one-way corridors. The Bellman optimality principle provides a natural framework for discrete control decisions, unifying anisotropic geometric guidance with directional constraints within a single computational framework.

Despite these advances, existing approaches often treat control parameters as fixed or manually tuned. The bi-level optimization of discrete channel configurations combined with continuous guidance intensities remains underexplored, particularly for large-scale urban pedestrian zones with multiple stages and route choices.

III. Macroscopic Crowd Dynamics Model

A. Hughes Model Foundation

We consider a bounded two-dimensional domain $\Omega \subset \mathbb{R}^2$ representing the pedestrian walkway. Let $\rho(x,t)$ denote the crowd density at position x and time t, $\phi(x,t)$ the minimal remaining time to reach an exit, and $\mathbf{v}(x,t)$ the mean velocity field. The continuity equation governing mass conservation reads:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0. \tag{1}$$

In the classical Hughes model [1], pedestrians move in the direction of steepest descent of the potential function:

$$\mathbf{v} = f(\rho)\frac{-\nabla\phi}{|\nabla\phi|}, \tag{2}$$

where $f(\rho)$ is the speed-density relationship. We adopt the Greenshields form:

$$f(\rho) = v_{\max}\left(1 - \frac{\rho}{\rho_{\max}}\right), \tag{3}$$

with $v_{\max}$ the free-flow speed and $\rho_{\max}$ the jam density.

The potential function satisfies the eikonal equation:

$$|\nabla\phi| = \frac{1}{f(\rho)}. \tag{4}$$

While this model captures basic congestion effects, it assumes isotropic movement and cannot represent channel geometries or directional controls. We address these limitations in the following sections.

IV. Channel Geometric Guidance via Metric Tensor

To represent anisotropic channel geometries such as corridors and barriers, we introduce a spatially varying symmetric positive-definite metric tensor $\mathbf{M}(x)$ that generalizes the classical eikonal equation to anisotropic form [3]:

$$\sqrt{\nabla\phi^\top \mathbf{M}(x)\nabla\phi} = \frac{1}{f(\rho)}. \tag{5}$$

Under this metric, the optimal direction is determined by the tensor-induced steepest descent:

$$\mathbf{v} = f(\rho)\frac{-\mathbf{M}(x)\nabla\phi}{\sqrt{\nabla\phi^\top \mathbf{M}(x)\nabla\phi}}. \tag{6}$$

For a channel region Ω_c with unit tangent $\tau(x)$ and unit normal $n(x)$, we construct the metric tensor as:

$$\mathbf{M}(x) = \alpha(x)\tau\tau^\top + \beta(x)nn^\top, \quad x \in \Omega_c, \tag{7}$$

where $\alpha(x)$ and $\beta(x)$ control the traversal weights in tangential and normal directions, respectively. Setting $\alpha(x) \gg \beta(x)$ encourages movement along the channel axis while suppressing lateral crossing, achieving continuous geometric guidance without explicit prohibition. Outside channels, $\mathbf{M}(x) = \mathbf{I}$ recovers the isotropic case.

Remark: The metric tensor approach represents directional preferences rather than hard constraints. As long as costs remain finite, the model allows individuals to deviate from preferred directions when facing congestion or detour requirements. This motivates the introduction of constrained HJB framework for strict directional controls.

V. Channel Control via Constrained HJB

Many crowd management measures directly restrict available movement directions rather than merely adjusting traversal costs. Examples include one-way corridors and temporary closures. Such rules are naturally modeled via constrained optimal control.

Let $U(x)$ denote the set of admissible unit directions at position x. The velocity satisfies:

$$\mathbf{v} = f(\rho)\mathbf{u}, \quad \mathbf{u}(x) \in U(x), \quad |\mathbf{u}| = 1. \tag{8}$$

By the Bellman optimality principle, the potential function satisfies the constrained HJB equation:

$$\max_{\mathbf{u}\in U(x)} \{-f(\rho)\mathbf{u}\cdot\nabla\phi\} = 1. \tag{9}$$

When $U(x)$ is the full unit circle, (9) reduces to the classical Hughes model. For restricted control sets, various channel rules can be expressed naturally. For a strict one-way channel with allowed direction $\tau(x)$:

$$U(x) = \{\tau(x)\}, \quad x \in \Omega_c, \tag{10}$$

and the HJB equation becomes:

$$-f(\rho)\tau(x)\cdot\nabla\phi = 1, \tag{11}$$

encoding one-way movement as a signed directional constraint.

A. Unified Discrete Bellman Formulation

To unify anisotropic geometry with discrete constraints, we employ a discrete Bellman formulation. For grid point x and admissible direction $\mathbf{u} \in U(x)$, define the step cost:

$$\text{Cost}(x, \mathbf{u}) = \frac{\Delta x}{f(\rho(x))} \frac{1}{\sqrt{\mathbf{u}^\top \mathbf{M}(x) \mathbf{u}}}, \tag{12}$$

The potential satisfies:

$$\phi(x) = \min_{\mathbf{u} \in U(x)} \left[\phi(x + \Delta x\, \mathbf{u}) + \text{Cost}(x, \mathbf{u}) \right]. \tag{13}$$

This formulation couples metric tensor $\mathbf{M}(x)$ for geometric guidance with control set $U(x)$ for directional constraints within a unified Bellman framework. The optimal direction is:

$$\mathbf{u}^*(x) = \arg \min_{\mathbf{u} \in U(x)} \left[\phi(x + \Delta x\, \mathbf{u}) + \text{Cost}(x, \mathbf{u}) \right], \tag{14}$$

with velocity $\mathbf{v}(x) = f(\rho) \mathbf{u}^*(x)$.

VI. Multi-Stage Multi-Route Model

Real-world scenarios often involve complex itineraries with multiple stages and route choices. We extend the density field to include stage and route indices, with fixed-probability splitting at decision boundaries.

A. Stage-Route Decomposition

Let the complete behavior chain consist of S stages, where stage s has R_s alternative routes. The total density decomposes as:

$$\rho(x, t) = \sum_{s=1}^{S} \sum_{r=1}^{R_s} \rho_{s,r}(x, t), \tag{15}$$

where $\rho_{s,r}(x, t)$ is the density of sub-population in stage s following route r.

All sub-populations share the same speed function $f(\rho)$ determined by total density, ensuring collective congestion effects. Sub-populations differ in their potential functions, optimal directions, and exit targets.

B. Sub-Population Potentials and Velocities

For each sub-population (s, r), define admissible direction set $U_{s,r}(x)$, metric tensor $\mathbf{M}_{s,r}(x)$, and target regions. The potential $\phi_{s,r}(x)$ satisfies:

$$\begin{aligned} \phi_{s,r}(x) = \min_{\mathbf{u} \in U_{s,r}(x)} \Big[& \phi_{s,r}(x + \Delta x\, \mathbf{u}) \\ & + \frac{\Delta x}{f(\rho)} \frac{1}{\sqrt{\mathbf{u}^\top \mathbf{M}_{s,r}(x) \mathbf{u}}} \Big]. \end{aligned} \tag{16}$$

The optimal direction is:

$$\begin{aligned} \mathbf{u}^*_{s,r}(x) = \arg \min_{\mathbf{u} \in U_{s,r}(x)} \Big[& \phi_{s,r}(x + \Delta x\, \mathbf{u}) \\ & + \frac{\Delta x}{f(\rho)} \frac{1}{\sqrt{\mathbf{u}^\top \mathbf{M}_{s,r}(x) \mathbf{u}}} \Big], \end{aligned} \tag{17}$$

with velocity $\mathbf{v}_{s,r}(x) = f(\rho)\, \mathbf{u}^*_{s,r}(x)$.

C. Stage Transitions and Fixed-Probability Splitting

Sub-populations couple through stage transitions. Let $G_{s,r} \subset \Omega$ be the completion goal region for stage (s, r) with indicator $\chi_{G_{s,r}}(x)$. Upon reaching this region, pedestrians switch to next-stage routes with fixed probabilities.

Let $p_{(s,r) \to q}$ be the fixed probability of switching from (s, r) to $(s + 1, q)$, satisfying:

$$p_{(s,r) \to q} \geq 0, \quad \sum_{q=1}^{R_{s+1}} p_{(s,r) \to q} = 1. \tag{18}$$

The transition term is:

$$Q_{(s,r) \to (s+1,q)}(x, t) = p_{(s,r) \to q}\, \kappa_{s,r}\, \chi_{G_{s,r}}(x)\, \rho_{s,r}(x, t), \tag{19}$$

where $\kappa_{s,r} > 0$ is the stage switching rate.

The density evolution for each sub-population includes advection and transition source terms:

$$\frac{\partial \rho_{s,r}}{\partial t} + \nabla \cdot (\rho_{s,r} \mathbf{v}_{s,r}) = Q^{\text{in}}_{s,r} - Q^{\text{out}}_{s,r}, \tag{20}$$

where $Q^{\text{in}}_{s,r}$ is inflow from the previous stage and $Q^{\text{out}}_{s,r}$ is outflow to the next stage.

VII. Numerical Solution

We discretize the domain using a uniform two-dimensional grid, with the control direction set represented as an 8-neighborhood basis with masking for regional restrictions.

Unified Bellman Solver. Rather than separate eikonal and HJB solvers, we use a unified discrete Bellman shortest-path update. For candidate direction $\mathbf{u}$, precompute geometric step factors from the local tensor, then combine with the speed function to form the complete step cost. A priority queue propagates potential values outward from exits until convergence. When the control set is unrestricted, this procedure is equivalent to monotone solution of the anisotropic eikonal equation; when controls are restricted, it naturally reduces to constrained HJB solution, covering all required scenarios within a unified framework.

Density Update. After obtaining converged potentials, recover velocity fields using (17), substitute into the continuity equation, and advance density explicitly using an upwind flux scheme:

$$\rho^{n+1} = \rho^n - \Delta t\, \nabla \cdot (\rho^n \mathbf{v}^n). \tag{21}$$

The time step satisfies the CFL stability condition.

Evaluation Oracle. For any given control configuration $z = (s, \eta)$, the simulation workflow constitutes a lower-level evaluation operator:

$$(\rho, \phi, \mathbf{v}) = \mathcal{S}(z; \hat{p}), \tag{22}$$

where $\hat{p}$ represents fixed route preference parameters identified from historical data. This operator provides the foundation for subsequent control optimization.

VIII. Control Evaluation Metrics

We distinguish between behavior validation metrics (for verifying model mechanisms) and control evaluation metrics (for assessing policy impacts). This section focuses on the latter.

A. Efficiency Metric: Total Travel Time

For simulation horizon $[0, T]$, the total travel time metric is:

$$J_1(z \mid \hat{p}) = \int_0^T \int_\Omega \rho(x,t)\, dx\, dt. \tag{23}$$

This measures aggregate dwell time of the crowd, providing a stable basis for comparing different control strategies.

B. Safety Metric: High-Density Exposure Time

Let ρ_{safe} be the safe density threshold. The exposure time metric is:

$$J_2(z \mid \hat{p}) = \int_0^T \int_\Omega \mathbf{1}_{\rho(x,t)>\rho_{\text{safe}}}\, dx\, dt. \tag{24}$$

This captures cumulative space-time exposure to dangerous crowding conditions, reflecting management concerns about sustained congestion risks.

C. Balance Metric: Channel Flow Variance

Let $q_c(t)$ be the flux through channel c at time t, with cumulative flow $F_c = \int_0^T q_c(t)\, dt$. The load balancing metric is:

$$J_5(z \mid \hat{p}) = \text{Var}(F_1, \ldots, F_m). \tag{25}$$

Lower variance indicates more even distribution of traffic across channels, reducing risk concentration and improving facility utilization.

D. Dimensionless Normalization

Because J_1, J_2, and J_5 have different physical units and numerical scales, we further define normalized indicators for policy comparison and optimization. Let $M_{\text{tot}} = M_{\text{init}} + M_{\text{in}}$ denote the total evaluated crowd mass, let $|\Omega_w|$ be the walkable area, and let $F = \sum_{c=1}^m F_c$ be the total cumulative channel flow. We define

$$\tilde{J}_1(z \mid \hat{p}) = \frac{J_1(z \mid \hat{p})}{M_{\text{tot}} T}, \tag{26}$$

$$\tilde{J}_2(z \mid \hat{p}) = \frac{J_2(z \mid \hat{p})}{|\Omega_w| T}, \tag{27}$$

and

$$\tilde{J}_5(z \mid \hat{p}) = \frac{\text{Var}(F_1, \ldots, F_m)}{F^2(m-1)/m^2} = \frac{m^2}{m-1}\text{Var}(p_1, \ldots, p_m), \tag{28}$$

where $p_c = F_c/F$. The normalized metrics are dimensionless, with $\tilde{J}_2 \in [0,1]$ and $\tilde{J}_5 \in [0,1]$.

E. Scalarized Optimization Objective

Given weights $(\lambda_1, \lambda_2, \lambda_5)$, the scalarized objective is:

$$J(z \mid \hat{p}) = \lambda_1 \tilde{J}_1(z \mid \hat{p}) + \lambda_2 \tilde{J}_2(z \mid \hat{p}) + \lambda_5 \tilde{J}_5(z \mid \hat{p}), \tag{29}$$

with the optimization problem:

$$\min_{z \in \mathcal{Z}} J(z \mid \hat{p}). \tag{30}$$

Our algorithm solves this scalarized problem for fixed weights, rather than directly generating the full Pareto frontier. Approximate Pareto sets can be obtained by running with different weight combinations.

IX. Optimization Method

A. Control Variable Definition

We distinguish controllable variables from exogenous behavioral parameters:

Controllable variables:

1) *Channel direction configuration* $s = (s_1, \ldots, s_C)$, where $s_c \in \{-1, 0, +1\}$ indicates reverse one-way, bidirectional, or forward one-way for channel c.
2) *Geometric guidance intensity* $\eta = (\eta_1, \ldots, \eta_C)$, where $\eta_c = \alpha_c/\beta_c \geq 1$ is the anisotropy ratio.

Thus the optimization variable is $z = (s, \eta)$.

Exogenous parameters: Route preference parameters $\hat{p}$ are identified from historical data and fixed during optimization. Decision region locations are reserved for future extension.

B. Structure-Aware Hybrid Block Optimization

Treating this as a black-box optimization ignores three key structures: mixed discrete-continuous variables, local Bellman response characteristics, and the expensive multi-group simulation structure.

We propose Structure-Aware Hybrid Block Optimization (SA-HBO), with four core components:

1) Discrete block neighborhood search for channel directions;
2) Surrogate pre-screening to reduce full simulations;
3) Bellman warm-start with local re-solving for efficient evaluation;
4) Continuous block projected stochastic approximation for guidance intensities.

C. Discrete Block Update

Fixing continuous variables η^k, the discrete subproblem at iteration k is:

$$s^{k+1/2} \approx \arg\min_{s \in \mathcal{S}} J(s, \eta^k \mid \hat{p}). \tag{31}$$

Define the neighborhood of current configuration s^k as:

$$\mathcal{N}(s^k) = \left\{ s : \|s - s^k\|_0 \leq \nu \right\}, \tag{32}$$

where ν is the maximum number of simultaneously flipped channels (typically 1 or 2).

Surrogate pre-screening: To reduce full simulations, construct a local surrogate for each candidate $s' \in \mathcal{N}(s^k)$:

$$\widehat{J}(s', \eta^k) = \mu_1 \widehat{J}_1^{\text{local}} + \mu_2 \widehat{J}_2^{\text{local}} + \mu_3 \widehat{J}_3^{\text{flow}}, \tag{33}$$

where the local metrics approximate density changes, exposure changes, and flow redistribution. Select top K candidates by surrogate value for full PDE simulation.

D. Continuous Block Update

Fixing discrete variables $s^{k+1/2}$, solve for η:

$$\eta^{k+1} \approx \arg\min_{\eta \in \mathcal{H}} J(s^{k+1/2}, \eta \mid \hat{p}), \tag{34}$$

with feasible set $\mathcal{H} = \{\eta : \eta_c \geq 1\}$.

Since gradients are unavailable analytically, we use projected stochastic approximation. Generate Rademacher perturbation $\Delta_\eta^k \in \{-1,+1\}^C$, construct two-sided perturbations $\eta_\pm^k = \eta^k \pm c_k \Delta_\eta^k$, and project to feasible set. The stochastic gradient estimate is:

$$\widehat{g}_\eta^k = \frac{J(s^{k+1/2}, \tilde{\eta}_+^k \mid \hat{p}) - J(s^{k+1/2}, \tilde{\eta}_-^k \mid \hat{p})}{2c_k} \odot (\Delta_\eta^k)^{-1}. \quad (35)$$

The candidate update is:

$$\eta_{\text{cand}}^{k+1} = \Pi_{\mathcal{H}} \left(\eta^k - \alpha_k D_\eta^{-1} \widehat{g}_\eta^k \right), \quad (36)$$

with diagonal preconditioner D_η.

Acceptance/Rejection: Compare candidate objective J_{cand} with current J_{cur}. Accept if $J_{\text{cand}} \leq J_{\text{cur}} - \varepsilon_{\text{acc}}$; otherwise reject and optionally backtrack. Maintain best solution z_{best} throughout.

E. Bellman Warm-Start and Local Re-solving

Control changes typically affect only local regions. We identify the affected region Ω_{chg}^k where $U_{s,r}(x;\tilde{z}) \neq U_{s,r}(x;z^k)$ or $\mathbf{M}_{s,r}(x;\tilde{z}) \neq \mathbf{M}_{s,r}(x;z^k)$. Using the old potential as initialization, iterate only within an expanded neighborhood $\Omega_{\text{chg}}^{k,\delta}$ until convergence.

This accelerates candidate evaluation. Periodic global corrections prevent error accumulation.

F. Algorithm Summary

1) Initialize potentials and density, compute $J(z^0 \mid \hat{p})$, set $z_{\text{best}} = z^0$.
2) For each iteration:
   - *Discrete update*: Generate neighborhood, screen with surrogate, evaluate top candidates with Bellman warm-start, select $s^{k+1/2}$.
   - *Continuous update*: Generate perturbations, compute stochastic gradient, generate candidate, accept/reject to obtain η^{k+1}.
   - *State update*: Set $z^{k+1} = (s^{k+1/2}, \eta^{k+1})$, compute objective, update z_{best} if improved.
3) Return $z^\star = z_{\text{best}}$.

Stopping criteria include: small objective change, no discrete variable change for consecutive iterations, or maximum iteration/simulation/time limits.

X. Numerical Experiments

This section presents numerical experiments to validate the model mechanisms, evaluate control strategies, and demonstrate the optimization framework. The experimental design follows a two-layer validation logic: first verifying whether the model mechanisms authentically alter local crowd behaviors, then comparing different channel control strategies in terms of efficiency, safety, and load balancing.

A. Experimental Setup

1) Three-Channel Scenario: The experiments employ a three-channel scenario as the primary testbed for mechanism validation and strategy comparison. The geometry consists of an initial crowd region on the left, an exit region on the right, and three passages (upper, middle, lower) through a central wall separating the two regions.

Numerical parameters are set as follows: grid resolution 96×72, spatial step $\Delta x = 0.5$, free-flow speed $v_{\max} = 1.5$, maximum density $\rho_{\max} = 5.0$, initial density $\rho_{\text{init}} = 2.2$, CFL coefficient 0.9, maximum simulation steps 600, and time horizon $T = 40$.

2) Evaluation Metrics: The experiments employ three primary raw metrics and their normalized counterparts:

- *Efficiency metric* J_1 and $\tilde{J}_1$: total travel time and its mass-time normalized form;
- *Safety metric* J_2 and $\tilde{J}_2$: high-density exposure time and its space-time normalized form;
- *Balance metric* J_5 and $\tilde{J}_5$: channel flow variance and its normalized imbalance index.

Behavioral indicators include channel capture domain ratios, flow distribution ratios, direction consistency indices, and approach angle distributions.

B. Mechanism Validation

The mechanism validation experiments address two fundamental questions: (1) Do geometric guidance $M(x)$ and directional constraints $U(x)$ authentically alter local decision-making and flow patterns? (2) Does the model produce sufficiently sensitive and interpretable responses to configuration changes?

1) Effect of Geometric Guidance $M(x)$: To verify whether geometric guidance alters approach patterns and attraction domains, we compare the baseline configuration (neither M nor U) against the M-only configuration (anisotropic guidance applied to the middle channel).

Results: Under the baseline configuration, the three channels exhibit approximately balanced capture domains: upper 33.73%, middle 30.00%, lower 34.98%. Flow distributions are similarly balanced: upper 33.82%, middle 32.35%, lower 33.83%.

Under the M-only configuration, the middle channel capture domain expands significantly to 63.41%, and its flow share jumps to 99.32%. This demonstrates that geometric guidance $M(x)$ reshapes local attraction domains and fundamentally alters approach patterns.

2) Effect of Directional Constraints $U(x)$: To verify whether directional constraints alter local optimal directions and channel selection, we compare the baseline against the U-only configuration (middle channel restricted to eastward-only flow).

Results: Under the U-only configuration, the middle channel capture domain drops to 0, and its flow share drops to 0, with upper and lower channels splitting traffic approximately $50/50$. This confirms that $U(x)$ significantly rewrites feasible path choices.

TABLE I
BIDIRECTIONAL $U(x)$ VALIDATION RESULTS

Configuration	Westbound Share	Counter-flow Mass	Head-on Exposure
Bidirectional baseline	73.19%	152.15	1055.74
Middle channel one-way	0	0	0

TABLE II
CONFIGURATION SENSITIVITY RESULTS

Config	Dominant Capture	Dominant Flow	Upper	Middle	Lower
middle	middle	middle	∼1/3	dominant	∼1/3
top	top	top	dominant	∼1/3	∼1/3
bottom	bottom	bottom	∼1/3	∼1/3	dominant

To further verify the suppression of counter-flow and head-on conflicts, we conduct a bidirectional flow sub-experiment. Two configurations are tested: bidirectional baseline (westbound and eastbound crowds simultaneously) and bidirectional with middle channel one-way rule.

The results in Table I show that when the middle channel enforces a one-way rule, westbound crowds are completely excluded, and counter-flow mass time and head-on cell exposure are both driven to zero. This confirms that $U(x)$ not only changes path selection but effectively suppresses counter-flow and head-on conflicts under bidirectional demand.

3) Combined $U + M$ Effects and Configuration Sensitivity: To verify whether the combined $U + M$ model produces distinguishable responses to different channel configurations, we test three setups: middle-channel-controlled ($U + M$-middle), upper-channel-controlled ($U + M$-top), and lower-channel-controlled ($U + M$-bottom).

Table II demonstrates that the $U + M$ model exhibits stable and interpretable responses to configuration changes: when the middle channel is controlled, it becomes the dominant capture and flow channel; when the upper or lower channel is controlled, the corresponding channel becomes dominant. This sensitivity validates the model as a foundation for subsequent strategy comparison and optimization.

4) Mechanism Validation Conclusions: The experimental results support four key conclusions:

1) Geometric guidance $M(x)$ significantly alters local approach patterns and channel attraction domains, enabling earlier alignment with channel axes and smoother convergence.
2) Directional constraints $U(x)$ reshape feasible paths and directional organization, effectively suppressing counter-flow and head-on conflicts under bidirectional demand.
3) The combined $U + M$ model produces configuration-sensitive responses that are both rule-compliant and behaviorally realistic.
4) Either U or M alone yields incomplete or behaviorally unrealistic responses; the full model is necessary for reliable optimization.

C. Control Strategy Comparison

This subsection compares different channel control strategies in terms of normalized efficiency ($\tilde{J}_1$), normalized safety ($\tilde{J}_2$), and normalized load balancing ($\tilde{J}_5$). The strategies under comparison include:

- Baseline (no control);
- Upper-channel-guided;
- Middle-channel-guided;
- Lower-channel-guided.

Detailed results and analysis will be presented in the full version of this paper.

D. Multi-Stage Flow Validation Experiments

This subsection validates the multi-stage behavior chain and probabilistic splitting mechanism. The experiments compare single-stage approximations against full multi-stage models to demonstrate how stage transitions and fixed-probability splitting alter density evolution and exit load distribution.

Detailed results and analysis will be presented in the full version of this paper.

E. Optimal Control Parameter Search

This subsection presents the first-version automated parameter search results. The search enumerates discrete strategy variables, performs coarse grid search on guidance intensity η, and searches over fixed splitting probability candidates to identify optimal control configurations under different objective weight combinations $(\lambda_1, \lambda_2, \lambda_5)$.

Detailed results and analysis will be presented in the full version of this paper.

XI. CONCLUSION

This paper presents a bi-level optimization framework for crowd channel management that combines anisotropic geometric guidance with discrete directional constraints. The proposed SA-HBO algorithm exploits problem structure through surrogate pre-screening, Bellman warm-start, and block coordinate updates to handle the expensive simulation-based optimization efficiently.

The numerical experiments validate that: (1) geometric guidance $M(x)$ and directional constraints $U(x)$ authentically alter local crowd behaviors; (2) the combined model exhibits stable sensitivity to configuration changes; (3) directional constraints effectively suppress counter-flow and head-on conflicts under bidirectional demand. These findings establish the foundation for subsequent strategy optimization studies.

Future work will extend to full multi-objective optimization, sensitivity analysis under demand perturbations, and calibration against real-world observation data.

REFERENCES

[1] R. L. Hughes, "A continuum theory for the flow of pedestrians," *Transportation Research Part B: Methodological*, vol. 36, no. 6, pp. 507–535, 2002.
[2] R. L. Hughes, "A continuum theory for the flow of pedestrians," *Transportation Research Part B: Methodological*, vol. 36, no. 6, pp. 507–535, 2002.
[3] E. Cristiani, F. S. Priuli, and A. Tosin, "Modeling rationality to control self-organization of crowds: An environmental approach," *SIAM Journal on Applied Mathematics*, vol. 75, no. 2, pp. 605–629, 2015.
[4] F. Priuli, A. Tosin, and E. Cristiani, "On the control of self-organized systems: The case of pedestrian flows," in *Proceedings of the 5th International Conference on Pedestrian and Evacuation Dynamics*, 2015.

[5] H. M. Al-Ahmadi *et al.*, "Preparedness for mass gatherings: A simulation-based framework for flow control and management using crowd monitoring data," *Arabian Journal for Science and Engineering*, vol. 46, no. 5, pp. 4985–4997, 2021.

[6] L. Feng and E. Miller-Hooks, "A network optimization-based approach for crowd management in large public gatherings," *Transportation Research Part C: Emerging Technologies*, vol. 42, pp. 182–199, 2014.

[7] R. X. Zhong *et al.*, "Boundary conditions and behavior of the macroscopic fundamental diagram based network traffic dynamics: A control systems perspective," *Transportation Research Part B: Methodological*, vol. 111, pp. 327–355, 2018.

[8] R. Zhao *et al.*, "Lyapunov-based crowd stability analysis for asymmetric pedestrian merging layout at T-shaped street junction," *IEEE Transactions on Intelligent Transportation Systems*, vol. 22, no. 11, pp. 6833–6842, 2021.

[9] S. A. Wadoo and P. Kachroo, "Feedback control of crowd evacuation in one dimension," *IEEE Transactions on Intelligent Transportation Systems*, vol. 11, no. 1, pp. 182–193, 2010.

[10] W. Qin, "Unit sliding mode control for disturbed crowd dynamics system based on integral barrier Lyapunov function," *IEEE Access*, vol. 8, pp. 91 257–91 264, 2020.

[11] Y. Zhu, D. Zhao, and H. He, "Optimal feedback control of pedestrian flow in heterogeneous corridors," *IEEE Transactions on Automation Science and Engineering*, vol. 18, no. 3, pp. 1097–1108, 2021.

[12] H. Yu *et al.*, "Regulating multi-directional passenger flow: Impact of obstacle position and flow level on pedestrian merging process," *Tunnelling and Underground Space Technology*, vol. 157, p. 106 336, 2025.

[13] Y. Zhou *et al.*, "Simulation-based adaptive optimization for passenger flow control measures at metro stations," *Simulation Modelling Practice and Theory*, vol. 138, p. 103 021, 2025.

[14] X. Yang *et al.*, "Optimization modeling of automatic crowd regulation at bottlenecks of subway system: A model predictive control approach," *Physics Letters A*, vol. 531, p. 130 180, 2025.

[15] X. Yang *et al.*, "Partition independent control and collaborative optimization of high-density crowd in subway stations," *Chaos, Solitons & Fractals*, vol. 200, p. 117 108, 2025.

[16] M. A. Lopez-Carmona and A. Paricio-Garcia, "Crowd evacuation management with optimal time-multiplexed exit-choice recommendations through simulation optimization," *Advanced Theory and Simulations*, vol. 7, no. 7, p. 2 400 265, 2024.

[17] M. A. Lopez-Carmona and A. Paricio Garcia, "Linear and nonlinear model predictive control (MPC) for regulating pedestrian flows with discrete speed instructions," *Physica A: Statistical Mechanics and its Applications*, vol. 625, p. 128 996, 2023.

[18] X. Liao *et al.*, "Crowd management through optimal layout of fences: An ant colony approach based on crowd simulation," *IEEE Transactions on Intelligent Transportation Systems*, vol. 24, no. 9, pp. 9137–9149, 2023.

[19] X. Liao *et al.*, "DRIFT: A dynamic crowd inflow control system using LSTM-based deep reinforcement learning," *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, vol. 55, no. 6, pp. 4202–4215, 2025.

[20] J. Schulman *et al.*, "Proximal policy optimization algorithms," *arXiv preprint arXiv:1707.06347*, 2017.

[21] K. Price, R. M. Storn, and J. A. Lampinen, *Differential Evolution: A Practical Approach to Global Optimization*. Springer Publishing Company, 2014.

[22] W. Liu *et al.*, "Automatic optimization of guidance guardrail layout based on multi-objective evolutionary algorithm," *Complex System Modeling and Simulation*, vol. 4, no. 4, pp. 353–367, 2024.